system, and using RubyInstaller[11] to install a Windows application that lets you execute Ruby programs.

The two different kinds of Ruby can both be installed on the same machine, and have different purposes. Using WSL gives you a command shell that is effectively a Linux distribution, allowing you to seamlessly use any of the other Ruby tooling in this book. Using RubyInstaller gives you access to Ruby from within a regular Windows PowerShell prompt, allowing you to execute Ruby programs from the File Explorer, and giving you access to Windows-specific libraries.

No matter which way you want to run Ruby, you should also install Windows Terminal so that you have a fully featured terminal program available. You can download Windows Terminal at https://docs.microsoft.com/en-us/windows/terminal/install, where there are also instructions on how to make it your default terminal program. From Windows Terminal, you can set up new command-line sessions using either Microsoft's PowerShell or the WSL shell. (You can also use Visual Studio Code's terminal to run either kind of command line.)

Also, if you need a brief tutorial on how Unix command lines work, see *Appendix 3, Command-Line Basics.*

Using Windows Subsystem for Linux

Windows Subsystem for Linux (WSL) allows you to run a Linux distribution binary inside your Windows setup without incurring the performance penalty of using a virtual machine or Docker container. WSL defines the wiring between the Linux OS commands and the Windows OS, allowing you to run your favorite Linux distribution from a command line transparently without having to deal with the Windows part at all. You can also have editing tools like Visual Studio Code or RubyMine interact with WSL as they run your Ruby code.

We're going to cover the basics of how to use WSL here, but if you need more information, the official documentation is available from Microsoft.[12]

Installing WSL

The first step in using Ruby with WSL is installing WSL itself. You need to be running Windows 11 or Windows 10 version 2004 or higher for this to work. We're installing WSL version 2 here.

You need to open an administrator Windows command terminal—it doesn't matter whether it's PowerShell or the regular terminal, but it does have to be an administrator shell. In Windows 11, the easiest way to get an admin shell is to right-click on the start menu and select the "Windows Terminal (Administrator)" option, which will open Windows Terminal in an admin shell. Depending on your Windows version, you may get prompted to say whether you will allow the program to make changes to your system. Say yes.

From the admin shell, type the command `wsl --install`. This will give us the default Linux installation, which is Ubuntu. The session looks like this when it's through, but it may take a little time to get through the download and installation process.

```
PS C: \Users\noelr> wsl --install
Installing: Virtual Machine Platform
```

11. https://rubyinstaller.org
12. https://docs.microsoft.com/en-us/windows/wsl